

ITA0612 – MACHINE LEARNING

CO4 – AT2

INDUSTRY PROBLEM SOLVING

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QUESTION: 1

A telecommunications company wants to recommend suitable data plans to customers based on their usage patterns. Design a solution using K-Nearest Neighbour or Case-Based Learning. Explain how similarity between users is measured, how recommendations are generated, and how the system adapts to changing user behavior over time.

ANSWER:

Telecom Company – Personalized Data Plan Recommendation

A telecommunications company can use **K-Nearest Neighbour (KNN)** or **Case-Based Reasoning (CBR)** to recommend suitable data plans based on customer usage patterns.

Proposed Solution

The system collects customer information such as:

- Monthly data usage
- Number of calls
- Number of SMS
- Average call duration
- Internet usage frequency
- Current plan
- Monthly bill amount

Each customer is represented as a feature vector.

For example: User A = [25 GB data, 300 calls, 50 SMS, ₹599 bill]

How similarity is measured

Using **KNN**, the similarity between two customers can be measured using **Euclidean distance**:

$$d = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

A smaller distance means that two customers have more similar usage patterns.

How recommendations are generated

1. Collect historical customer usage data.
2. Convert customer information into numerical features.
3. Calculate the distance between the new customer and existing customers.
4. Select the **K most similar customers**.
5. Analyse the plans used by those customers.
6. Recommend the plan most frequently selected by the nearest customers.

Example:

If 5 similar customers are found:

- 3 use 50 GB plan
- 1 uses 30 GB plan
- 1 uses 75 GB plan

The system recommends the **50 GB plan**.

How the system adapts over time

The recommendation system can continuously collect new usage information. When customers change their plans or their usage patterns change, the new data is added to the historical dataset.

Therefore, recommendations can be updated periodically based on recent customer behaviour.

Case-Based Reasoning

In Case-Based Reasoning, the system stores previous customer cases along with the plans recommended to them.

For a new customer: **Retrieve** → **Compare** → **Reuse** → **Adapt**

The system finds a previous customer with a similar usage pattern and adapts the previous recommendation to the new customer.

Conclusion: KNN/CBR provides personalized recommendations by comparing a customer's current behaviour with similar historical customers. Continuous collection of new usage data allows the system to improve recommendations over time.

QUESTION: 2

An energy management company aims to predict electricity consumption based on factors such as time, weather, and appliance usage. Propose a solution using Locally Weighted Regression or RBF networks. Explain how local learning captures variations in consumption patterns and how the model handles noisy and seasonal data.

ANSWER:

Energy Management – Electricity Consumption Prediction

An energy management company can use Locally Weighted Regression (LWR) or an RBF (Radial Basis Function) network to predict electricity consumption.

The input features can include:

- Time of day
- Day of week
- Temperature
- Humidity
- Weather conditions
- Number of appliances being used
- Previous electricity consumption

The target output is: **Predicted electricity consumption**

Proposed Solution Using Locally Weighted Regression

Unlike ordinary regression, Locally Weighted Regression gives greater importance to data points that are close to the current input.

For example, when predicting electricity consumption at **7 PM**, consumption records from nearby times such as 6 PM and 8 PM receive higher weights than records from 3 AM.

How local learning works

Suppose the company wants to predict consumption for:

Time = 7 PM, Temperature = 32°C

The algorithm searches for historical observations with similar conditions.

Nearby observations receive higher weights:

Historical Condition	Similarity	Weight
6:30 PM, 31°C	High	High
7:00 PM, 32°C	Very High	Very High
7:30 PM, 33°C	High	High
2:00 AM, 25°C	Low	Low

The prediction is therefore mainly influenced by the local observations.

Why local learning is useful

Electricity consumption can change depending on:

- Morning/evening usage
- Weekdays/weekends
- Summer/winter
- Temperature
- Appliance usage
- Holidays

A single global regression model may not capture all these variations effectively.

LWR builds the prediction around the neighbourhood of the current input, allowing it to capture local patterns and seasonal behaviour.

Handling Noisy Data

Real-world electricity data may contain unusual values due to:

- Power failures
- Sudden appliance usage
- Sensor errors
- Abnormal weather
- Unexpected events

The model can reduce the influence of distant or less relevant observations by assigning them lower weights.

Data preprocessing can also be performed using:

- Missing-value handling
- Outlier detection
- Normalization
- Noise filtering

Handling Seasonal Data

Electricity consumption can be different during different seasons.

For example:

Summer: High consumption because of air conditioners and fans.

Winter: Different consumption due to heating requirements.

Weekends: Consumption patterns may differ from weekdays.

LWR can capture these changes because predictions are based on nearby and similar historical observations.

RBF Network Alternative

An RBF neural network can also be used for electricity prediction.

An RBF network contains:

1. **Input layer**
2. **Radial basis function/hidden layer**
3. **Output layer**

The RBF function measures how close an input is to a particular centre.

A commonly used Gaussian RBF is:

$$\phi(x) = e^{-(2\sigma^2 \|x - c\|^2)}$$

where:

- x = input
- c = centre
- σ = width of the function

Similar electricity-consumption patterns produce stronger RBF responses.

Advantages of RBF

- Captures nonlinear relationships.
- Learns local patterns.
- Works well with changing consumption behaviour.
- Can model complex relationships between weather, time and appliance usage.

Final comparison

Problem	ML Technique	Main Idea
Telecom plan recommendation	KNN / CBR	Find similar customers and recommend their suitable plans
Electricity prediction	Locally Weighted Regression	Give more importance to locally similar observations
Electricity prediction alternative	RBF Network	Learn nonlinear local patterns using radial basis functions